

Uwais Alqarni

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EDUCATION

Singapore Institute of Technology <i>BEng (Hons) in Information and Communications Technology, Software Engineering</i>	Singapore 2024 – Present
Nanyang Polytechnic <i>Diploma in Information Technology</i>	Singapore 2019 – 2021
• Distinctions: UX Design, Networking Technology; Director's List (Top 15% of cohort), 2021 & 2022 • Google Data Analytics Specialization, Coursera – 2022	

PUBLICATIONS

- Derick C.Z. Lee, **Uwais Alqarni Bin Abdul Rahman**, Yu-Xuan Tan, Yi-Zhen Cheng, Benjamin Wj Kwok, Kan Chen, and Jeannie S.A. Lee. 2026. *TakeoverBench: A Benchmarking Platform for Takeover Requests (TOR) in Conditionally Automated Vehicles*. In Adjunct Proceedings of the 18th International Conference on Automotive User Interfaces and Interactive Vehicular Applications (AutomotiveUI Adjunct '26), Gothenburg, Sweden. ACM. doi.org/10.1145/3828158.3833210
- 2nd author of 7 – built the system's Flask backend and core simulation baseline (later extended by teammates), and contributed to researching and selecting the takeover-performance metrics used for the benchmark, alongside general team research
- Yi-Zhen Cheng, Yu-Xuan Tan, **Uwais Alqarni Bin Abdul Rahman**, Derick C.Z. Lee, Kan Chen, and Jeannie S.A. Lee. 2026. *Student Research Track: Towards Low-Cost Driver Supervision: Evaluating a Smartphone Interface for Driver Inattention Monitoring in Simulated Driving*. In Adjunct Proceedings of the 18th International Conference on Automotive User Interfaces and Interactive Vehicular Applications (AutomotiveUI Adjunct '26), Gothenburg, Sweden. ACM. doi.org/10.1145/3828158.3834806
- 3rd author of 6 – developed the custom YOLOv8 object-detection model used to classify driver non-driving-related tasks (phone-like interaction, reading), and contributed to general research and the final demo

EXPERIENCE

Software Engineer <i>Aktus MU Kreatif</i>	Sep 2021 – Nov 2021 Singapore
• Applied object-oriented design principles to implement front-end and back-end updates to a customer-facing Progressive Web App, adding offline functionality with reliable, well-tested data synchronization • Built and shipped a machine learning feature end-to-end using Python to forecast product pricing, driving a 15% increase in website revenue • Used Git for version control across all code changes, collaborating with the product team to scope requirements and communicate technical trade-offs	

PROJECTS

Pergas Members Integrated System <i>Next.js, React, Supabase (PostgreSQL), Playwright</i>	2026
• Partnered with Pergas, a registered Singapore NGO, on a real-client project, collaborating with stakeholders to translate requirements into a unified Member and Admin web portal spanning the full development life cycle from requirements gathering to deployment • Architected a full-stack Member and Admin portal using Next.js (App Router, SSR) with server-side Role-Based Access Control, and integrated third-party services (Zoho Backstage, Google Maps) through RESTful API design • Designed and deployed a relational Supabase (PostgreSQL) schema spanning 10+ interconnected tables to enforce data integrity across member tiers, event registrations, and partner perks • Authored a Playwright test suite achieving a 100% pass rate across 20 regression scenarios covering API security, file validation, and accessibility; implemented server-side data-privacy measures (e.g., scrubbing NRIC) for PDPA compliance, and coordinated delivery across bi-weekly Agile sprints via a GitHub Projects Kanban board	
Clean Brilliant Company – Carbon Analytics Platform <i>C#/.NET, ASP.NET Core, PostgreSQL</i>	2025

- Built the Dashboard Customization & Layout Persistence Engine for an enterprise C#/.NET web application, implementing the Table Data Gateway pattern with asynchronous, parameterized ADO.NET/Npgsql queries against PostgreSQL to prevent SQL injection
- Designed a hybrid relational + JSONB schema to persist flexible, schema-less dashboard layouts, and integrated Unit of Work to track state transitions across create/update/delete operations
- Developed complete layout lifecycle logic (add, remove, reorder, auto-pack widgets), applying Strategy, Builder, and Serialization design patterns within a Boundary-Control-Entity architecture wired through ASP.NET Core dependency injection
- Built MVC controller endpoints including a drag-and-drop reordering API consuming dynamic JSON payloads, and collaborated with a multi-person team via Git feature-branch workflows and pull requests

Codex – AI Video Knowledge Engine | *Python, FastAPI, Gemini API, PostgreSQL, AWS S3* 2024

- Architected a multimodal Retrieval-Augmented Generation (RAG) pipeline using the Gemini API and pgvector embeddings, enabling semantic search and Q&A-style insight extraction across video, audio, and visual content
- Built and tuned the retrieval pipeline (chunking, embedding, vector search) through iterative experimentation to improve semantic-search relevance
- Developed a fault-tolerant, asynchronous FastAPI backend with lazy database connections to eliminate timeouts under high-load processing, deployed with AWS S3 for cloud storage, and cut database infrastructure costs by 30% through data-driven optimization

BidForGood – Secure Charity Auction Platform | *React, TypeScript, Node.js, PostgreSQL, GitHub Actions* 2024

- Collaborated with a 4-person team on a full-stack application, contributing front-end and back-end code through pull requests and structured code reviews as part of a secure-software-development module
- Designed and built RESTful backend endpoints on a relational PostgreSQL database covering authentication, bidding, and input validation, and wrote automated tests using Node's built-in test runner
- Built and maintained a CI/CD pipeline in GitHub Actions running dependency installation, secret scanning, automated tests, and dependency-vulnerability audits on every push, and implemented secure-session and CSRF-protection features

Tabayyun – Hybrid AI Food Scanner | *React, TensorFlow.js, YOLOv8n, Python* 2023

- Built a hybrid computer-vision pipeline combining a YOLOv8n object-detection model with Tesseract.js OCR, compiled to WebAssembly for real-time, on-device inference in an offline-first Progressive Web App
- Optimized model inference performance on mobile hardware through dynamic resolution scaling, reducing RAM overhead and iOS crash rates
- Designed and evaluated a custom heuristic post-processing algorithm to correct OCR errors, iteratively testing against ground-truth labels to achieve 90% accuracy

TECHNICAL SKILLS

Languages: C# (.NET, ASP.NET Core), Java, Python, SQL, TypeScript/JavaScript

Software Concepts: OOP, design patterns (Table Data Gateway, Unit of Work, Strategy, Builder, BCE), REST APIs, data structures & algorithms

Frontend: HTML5, CSS3, React, Next.js, Vue.js, Tailwind CSS

Backend: Node.js/Express, FastAPI, ASP.NET Core, RESTful API design

ML & GenAI: Gemini API, Retrieval-Augmented Generation (RAG), prompt engineering, TensorFlow.js, YOLOv8n (object detection), OCR (Tesseract.js)

Databases: PostgreSQL, Supabase, ADO.NET/Npgsql, Entity Framework Core, JSONB, pgvector, ChromaDB (NoSQL/vector)

Cloud & DevOps: AWS S3, Docker, Git/GitHub, GitHub Actions (CI/CD)

Practices: Agile/Scrum (sprints, backlog grooming, Kanban), automated testing, code review, technical documentation